

# NIKITA CHAURASIA

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [LeetCode](#) | [GeeksforGeeks](#)

## TECHNICAL SKILLS

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**Programming:** Python, C++, SQL

**CS Subjects:** Data Structures & Algorithms (DSA), Operating Systems, Database Management Systems (DBMS), Computer Networks, Object Oriented Programming (OOP)

**AI/ML:** Machine Learning, Supervised Learning, Feature Engineering, Model Training, LightGBM

**Problem-Solving:** Algorithm Design, Complexity Analysis, Debugging & Optimization, System Design

**Analytical Skills:** Logical Reasoning, Pattern Recognition, Data Interpretation

## SUMMARY

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- Results-driven Software Engineer with a strong foundation in Electronics, AI/ML, and Data Structures and Algorithms. Skilled in building secure, scalable applications and applying machine learning to real-world data pipelines, with a passion for system design, efficient algorithms, and software that performs reliably under load.

## PROJECTS

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**SENTINEL** – AI Prompt Security Gateway

June 2026

*Python, Chrome Extension (Manifest V3), React, Chart.js, FastAPI, JWT*

- Built a Manifest V3 Chrome extension that intercepts prompts on ChatGPT, Claude, and Gemini, scanning for API keys, credentials, and PII before submission with zero network calls in the detection path.
- Implemented 5 algorithms from scratch — Aho-Corasick, Shannon entropy, Trie, Max-Heap, and Merge-Intervals — detecting 24 secret patterns plus base64-encoded payloads via a decode-and-rescan layer.
- Shipped a React + Chart.js dashboard backed by a FastAPI + JWT service, 3 tunable sensitivity levels, and 104 tests covering ReDoS safety and a false-positive corpus.

Live Demo: [nikitayk.github.io/SENTINEL](https://nikitayk.github.io/SENTINEL) | GitHub: [github.com/nikitayk/SENTINEL](https://github.com/nikitayk/SENTINEL)

**ADPULSE** – Real-Time Demand-Side Platform

June 2026

*Python, LightGBM, Flask-SocketIO, Three.js*

- Built a full-stack Demand-Side Platform (DSP) that participates in real-time ad auctions, prices each bid in-memory at ~5 ms, and sustains a live feed of 7 bids/sec.
- Trained 2 LightGBM models (CTR + CVR) on IPinYou competition logs with 12 engineered features; bid pricing scales up to 11x the base price with a budget pacer and a hard cap of 300.
- Engineered a generator-based pipeline that streams multi-GB logs in O(1) memory; a Flask-SocketIO + Three.js 3D dashboard renders live KPIs, validated across a Python 3.9/3.11 CI matrix with 46 tests.

Deployed Link: [adpulse-457r.onrender.com](https://adpulse-457r.onrender.com) | GitHub: [github.com/nikitayk/ADPULSE](https://github.com/nikitayk/ADPULSE)

## EDUCATION

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**B.Tech in ECAM (ECE + AI/ML)**

2023 – 2027

Netaji Subhas University of Technology — 6.88 CGPA

**Senior Secondary School (CBSE Class XII)**

2023

Rani Jhansi Sarvodaya Kanya Vidhyalaya, New Delhi — 83%

**Secondary School (CBSE Class X)**

2021

Rani Jhansi Sarvodaya Kanya Vidhyalaya, New Delhi — 95.5%

## CERTIFICATIONS

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- [AI For Everyone](#) — DeepLearning.AI
- [Supervised Machine Learning \(Regression and Classification\)](#) — Stanford Online
- [Advanced Learning Algorithms](#) — DeepLearning.AI

## ACHIEVEMENTS / EXTRACURRICULAR

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- Solved 300+ Data Structures and Algorithms problems on LeetCode and GeeksforGeeks, demonstrating consistent problem-solving rigor.
- Ranked within the top 4 percentile among 2,500+ teams at DTU Execute 4.0 Hackathon for technical execution under time constraints.
- Secured a Top 20 finish at Hackmania (Microsoft Azure) and The Elites (Microsoft Gurugram), competing on cloud and enterprise-track challenges.